PACMAN Quality Assurance and Quality Control

Hardware checkout: tools, coverage, and shifter's guide

Draft maintained by the PACMAN agents (augie); reference for the TRM

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1 Purpose and scope

- Define a repeatable checkout of every new PACMAN card before it goes into a production system or onto an ASIC test stand.
- Two utilities, both run on the PACMAN's own Linux system:
  - `pacman_lnxcheck`: the standard embedded-Linux checkout of the Trenz module (CPU, memory, on-chip supplies, Ethernet, flash and SD storage).
  - `pacman_hwcheck`: the PACMAN-specific checkout (power, tile I/O, timing inputs, front panel), partly guided by an operator.
- Scope: the features needed for normal operation. Features that normal operation does not use are listed (Section 6) but not tested.
- Method: wherever possible a feature is tested by loopback, so that the feature and its feedback path are verified together, and a failure points at hardware, firmware or software.

2 Production setting

- PACMAN 1v5 with a Trenz TE0720 module (Zynq-7000). The card boots from its microSD card (boot image and Linux root file system) and is reached over Ethernet only (`ssh`). Production cards have no USB/JTAG header installed.
- **ASIC protection jumper (J5).** Installed whenever the card is in the production system or connected to ASICs. The checkout runs with J5 *removed*; with J5 installed, `pacman_hwcheck` refuses everything except its read-only report.
- The PACMAN data and command servers are stopped during the checkout (they share the data path with the loopback test) and restarted afterwards.

3 Test setup

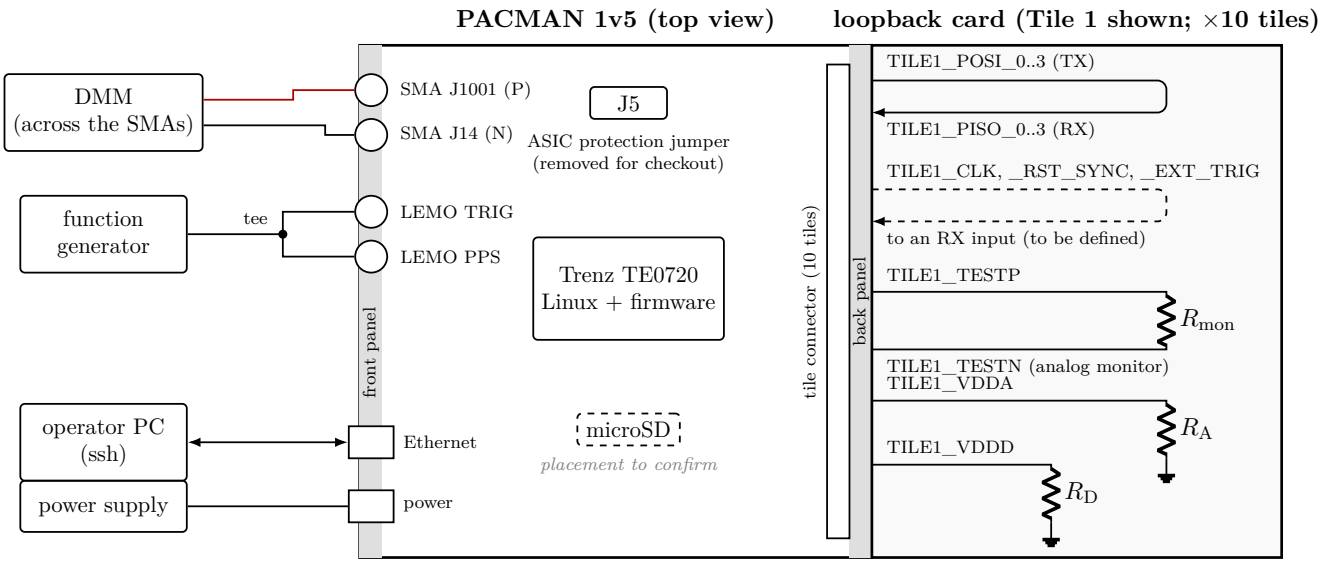


Figure 1: Checkout setup (physical view; connector placement to be confirmed). All loopbacks are on the loopback card; Tile 1's connections are shown with their signal names, and the card repeats them for all ten tiles. Dashed: routing still to be defined.

Item	Role
Loopback card	On the back-panel tile connector, for all ten tiles: each tile's data outputs (POSI) loop back to its data inputs (PISO); its clock, sync and trigger outputs loop back to an RX input; a resistor closes its analog monitor pair (TESTP/TESTN); load resistors on VDDA and VDDD let the tile supplies be swept under load.
DMM	Front-panel SMA readings, entered by the operator when prompted.
Function generator	One output, split with a tee: the same square wave on the TRIG and PPS LEMO inputs.
Operator PC	`ssh` session to the PACMAN; keeps the result files.

- Two versions of the loopback card are needed for different tests, so the checkout is split into one run per card, with the PACMAN powered down while the cards are swapped. Which tests go on which card: [TBD: lab day].

4 The utilities

4.1 Common conventions

- One line per result: `<VERDICT> <name>-- <value>`, then a final `VERDICT` line.
- Result verdicts: `PASS`, `FAIL`, `SKIP` (not done, with the reason), `INFO` (recorded, not judged). `pacman_hwcheck` also prints `DATA` lines (sweeps).
- Run verdict and exit code:

Run verdict	Meaning	Exit code
`PASS`	every check passed, nothing skipped	0
`FAIL`	at least one check failed	1
`INCOMPLETE`	no failure, but something was skipped	2
(refused / usage)	bad options, or J5 installed	3

- Only a `PASS` run qualifies a card.

4.2 `pacman_lnxcheck`: embedded Linux

Section	Checks
System	CPU count, memory size, kernel version.
On-chip monitors	Die temperature; six Zynq supplies against nominal.
Ethernet	Link speed and duplex (1000/full), MAC address, IP address; ping with small and full-size packets, no loss; TFTP transfer of a file twice, compared; interface error counters unchanged.
QSPI flash	Each partition read three times; the last two reads must agree. A difference on the first read after boot is reported (INFO), not failed.
eMMC	First 64 MB read twice, compared; read rate reported.
microSD	Card present, partitions, boot image checksum. Optional `--sd_write`: a 16 MB test file written, read back and removed.

Read-only unless `--sd_write` is given. Options: `--server` (TFTP server; default the default gateway), `--tftp_file`, `--no_tftp`, `--sd_write`.

4.3 `pacman_hwcheck`: PACMAN hardware

Run sets execute in this fixed order; `--full_checkout` runs all of them and stops at the first failing section (`--continue` keeps going).

Run set	Checks	Operator
`--read_only`	Firmware and hardware versions, build information, J5 state, board and tile monitors, data path status. Allowed with J5 installed.	–
`--blink`	Trenz and PACMAN LEDs blink in turn.	watches, confirms
`--power`	Board rails against nominal; tile power off, then on. Each tile in turn: VDDA and VDDD swept over the DAC range into the card's loads (floor, gain, swing, monotonic, load), other tiles stay off. Board-rail test points surveyed with the DMM.	DMM readings
`--loopback`	Each tile in turn, only that tile enabled: random data on its four UARTs through the loopback card, words received, bit errors, receiver error counters; no stray data on other UARTs.	–
`--timing_inputs`	Rates on the TRIG and PPS inputs, counted by the firmware over 1 s.	–
`--analog_io`	Front-panel analog MUX stepped through tiles 1–10, test DAC, off; a DMM reading at the SMA jacks for each.	DMM readings

- Operator steps start with a prompt saying what comes next (`Ready?  [y = go, s = skip]`). At any prompt, `s` skips the step; it is recorded as SKIP, never as a value.

- Modifiers: `--tile_mask` (tiles present on the fixture), `--skip_interactive` (no operator steps), `--continue`; development: `--internal_loopback`, `--no_i2c`, `--sim_i2c`, `--ber_bits`, `--analog_io_mode`.

5 Coverage and acceptance limits

Feature	Acceptance	Status
Board rails 3V6, 3V3, 3V0	monitor within 5% of nominal; DMM within 5% of monitor; 3V0 $\leq$ 50 mV when off	limits set; DMM test points [TBD:]
Tile supplies VDDA, VDDD	floor and gain (157 mV per 0x1000 DAC codes, $\pm$5%), swing $\geq$ 1 V, monotonic, current under load, other tiles $\leq$ 50 mV	provisional (one board)
Tile UARTs (40)	every word received, zero bit errors, no stray words	set
TRIG, PPS inputs	rate matches the generator	rates recorded; limits [TBD:]
Front-panel analog MUX	expected SMA readings	readings recorded; values [TBD:]
LEDs	operator confirms	set
Linux, Ethernet, storage	as in the `pacman_lnxcheck` table	set

6 Status and untested features

Needed for normal operation, not yet tested	Plan
Tile CLK, TRIG, SYNC outputs	Loopback to an RX input on a separate connector (fixture [TBD:]).
Analog monitor	Known resistor across the input, measured through the front panel.
TRIG / PPS acceptance	Function generator at the lab; then rate limits.
Full ten-tile run	Loopback card covering all ten tiles.
microSD checks	First run on an SD-booted card.

Not needed for normal operation (not tested): onboard ADC, RTD temperature probe, QSPI flash contents, SFP/CDR timing path.

7 Shifter's guide: hardware checkout

For an operator who runs the checkout the standard way. Items marked [TBD: ...] are still to be filled in.

You need

- The PACMAN card with its microSD card (production image), and its serial number [TBD: where printed].
- Loopback / load card, DMM, function generator with two outputs, Ethernet cable, PACMAN power supply.

- A PC on the same network, with `ssh`.

Prepare

1. Power off. Remove J5 (ASIC protection jumper). Make sure no ASICs are connected.
2. Install the loopback / load card on the tile connectors.
3. Connect the function generator to the TRIG and PPS LEMO inputs: square wave, [TBD: amplitude], [TBD: frequency].
4. Connect Ethernet; power on. Wait about [TBD: N] s for Linux to boot.
5. Log in: `ssh [TBD: user]@[TBD: PACMAN address]`.
6. Stop the PACMAN servers: `/etc/init.d/pacman_server stop`

Run

1. Linux checkout (a minute or less):
   `pacman_lnxcheck --server [TBD: TFTP server] | tee lnxcheck-<serial>.txt`
2. Hardware checkout (guided; about [TBD: N] minutes):
   `pacman_hwcheck --full_checkout | tee hwcheck-<serial>.txt`
3. Answer the prompts:
   - `Next: ... Ready?`: get ready for what it describes (watch the LEDs, put the DMM probes on the named points), then answer `y`.
   - Yes/no questions: answer what you saw.
   - Readings: type the DMM value in mV.
   - Can't do a step? Answer `s`. It is recorded as skipped; never type a made-up value.

Read the result

- The last line of each run is the verdict.
- `PASS` from both: the card passes.
- `FAIL`: the lines marked `FAIL` say what failed. The hardware checkout stops at the first failing section. Do not repeat blindly: record, and report to [TBD: contact].
- `INCOMPLETE`: something was skipped (lines marked `SKIP` give the reason). Fix the setup and run again.
- `REFUSED` with J5 mentioned: J5 is installed; power off, remove it, start again.

Finish

1. Copy both result files to [TBD: results location], named by serial number.
2. Restart the servers (`/etc/init.d/pacman_server start`) or reboot.
3. Power off; remove the fixtures. Install J5 before the card goes to a production system or ASICs.
4. Record the card's status in [TBD: tracking sheet].